EXPERIMENTS

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COURSE CODE:**DSA0601 DATA HANDLING ANDVISUALIZATION**

SET 2 – WEBSITE ANALYTICS

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

QUESTION 1

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

AIM

To create a line chart to visualize the trend of daily website page views over time.

PROCEDURE

1. Create the website traffic dataset.
2. Store the date and page views.
3. Convert the date into Date format.
4. Use the plot() function to create a line chart.
5. Label the X-axis and Y-axis.
6. Add a suitable title.

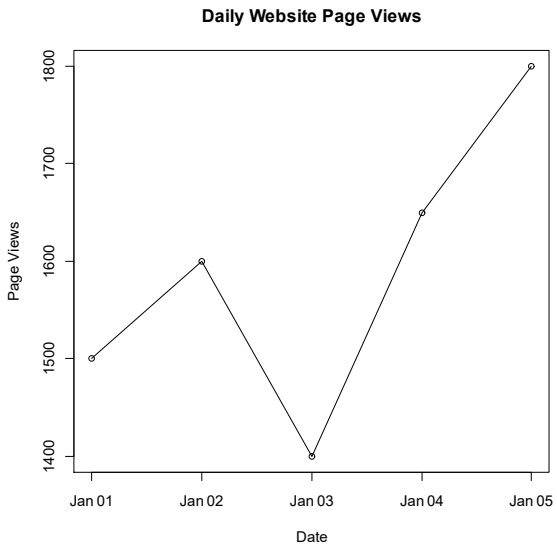
CODE

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"))

page_views <- c(1500, 1600, 1400, 1650, 1800)

plot(
```

```
  date,
  page_views,
  type = "o",
  xlab = "Date",
  ylab = "Page Views",
  main = "Daily Website Page Views"
)
```**OUTPUT**



RESULT

Thus, the line chart was successfully created to visualize the daily trend in website page views.

QUESTION 2

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

AIM

To create a bar chart showing the days with the highest click-through rates.

PROCEDURE

1. Create the date and click-through-rate data.
2. Store the values in a data frame.
3. Sort the click-through rates in descending order.
4. Select the top N days.
5. Create a bar chart using barplot().
6. Add labels and title.

CODE

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")

ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

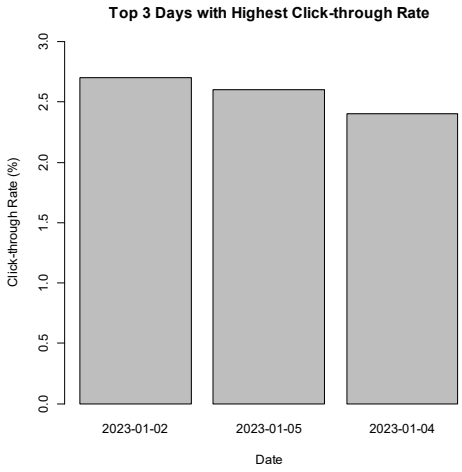
website <- data.frame(
  Date = date,
  CTR = ctr
)

website <- website[order(-website$CTR), ]

top_n <- website[1:3, ]

barplot(
  top_n$CTR,
  names.arg = top_n$Date,
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  main = "Top 3 Days with Highest Click-through Rate",
  ylim = c(0, 3)
)
```

OUTPUT



RESULT

Thus, the bar chart was successfully created to display the top 3 days with the highest click-through rates.

QUESTION 3

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

AIM

To create a stacked area chart showing the distribution of likes, shares, and comments over time.

PROCEDURE

1. Create the date data.
2. Enter likes, shares, and comments.
3. Create a data frame.
4. Use the stack() function to organize the interaction data.
5. Use matplot() to create the stacked area chart.
6. Add labels, title, and legend.

CODE

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"))

likes <- c(120, 150, 100, 170, 200)
shares <- c(40, 55, 35, 60, 75)
comments <- c(20, 25, 18, 30, 35)

interactions <- cbind(likes, shares, comments)

cumulative <- apply(interactions, 1, cumsum)

plot(
  date,
  cumulative[1, ],
  type = "n",
  ylim = c(0, max(cumulative)),
  xlab = "Date",
  ylab = "Number of Interactions",
  main = "Website User Interactions"
)

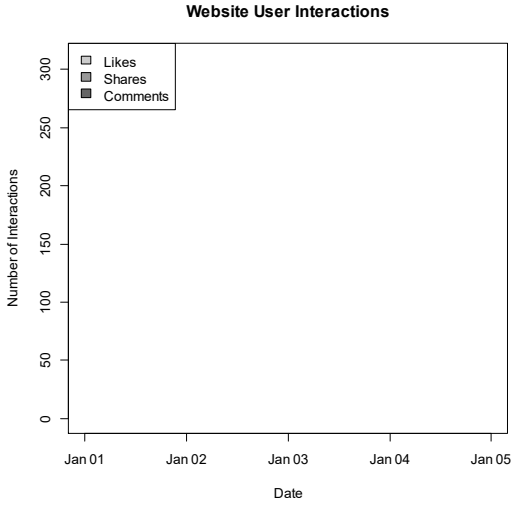
polygon(
  c(date, rev(date)),
  c(rep(0, length(date)), rev(cumulative[1, ])),
  border = NA
)

polygon(
```

```
  c(date, rev(date)),
  c(cumulative[1, ], rev(cumulative[2, ])),
  border = NA
)

polygon(
  c(date, rev(date)),
  c(cumulative[2, ], rev(cumulative[3, ])),
  border = NA
)

legend(
  "topleft",
  legend = c("Likes", "Shares", "Comments"),
  fill = c("grey80", "grey60", "grey40")
)
```
OUTPUT



RESULT

Thus, a stacked area chart was successfully created to visualize the distribution of website user interactions.

QUESTION 4

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

AIM

To develop an interactive website traffic dashboard using R Shiny with a line chart and bar chart.

PROCEDURE

1. Install and load the shiny and ggplot2 packages.
2. Create the website traffic dataset.
3. Create a user interface with a date selection.
4. Create a line chart for page views.
5. Create a bar chart for click-through rates.
6. Display both charts in the dashboard.
7. Allow the user to select dates interactively.
8. Run the Shiny application.

CODE

```
library(shiny)
library(ggplot2)

website <- data.frame(
  Date = as.Date(c(
    "2023-01-01",
    "2023-01-02",
    "2023-01-03",
    "2023-01-04",
    "2023-01-05"
  )),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),

  sidebarLayout(
    sidebarPanel(
      dateRangeInput(
        "date",
        "Select Date Range",
        start = min(website$Date),
        end = max(website$Date)
      )
    ),
```

```
    mainPanel(
      plotOutput("lineChart"),
      plotOutput("barChart")
    )
  )
)

server <- function(input, output) {

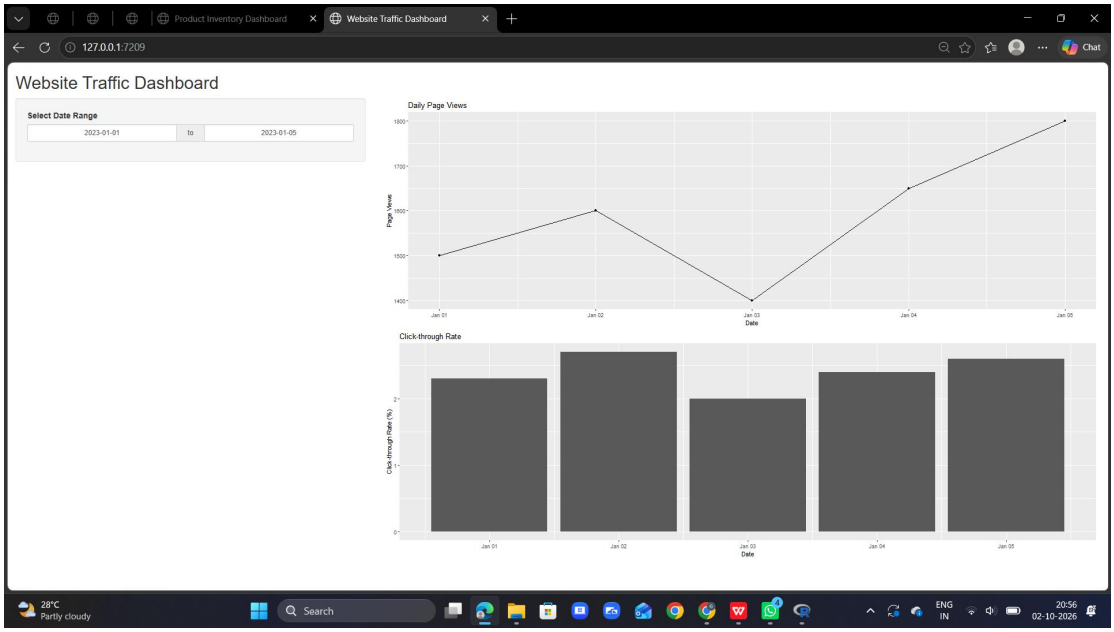
  filtered_data <- reactive({
    website[
      website$Date >= input$date[1] &
      website$Date <= input$date[2],
    ]
  })

  output$lineChart <- renderPlot({
    ggplot(
      filtered_data(),
      aes(x = Date, y = Page_Views)
    ) +
      geom_line() +
      geom_point() +
      labs(
        title = "Daily Page Views",
        x = "Date",
        y = "Page Views"
      )
  })

  output$barChart <- renderPlot({
    ggplot(
      filtered_data(),
      aes(x = Date, y = CTR)
    ) +
      geom_bar(stat = "identity") +
      labs(
        title = "Click-through Rate",
        x = "Date",
        y = "Click-through Rate (%)"
      )
  })
}

shinyApp(ui = ui, server = server)
```

OUTPUT



RESULT

Thus, an interactive **Website Traffic Dashboard** was successfully developed using R Shiny with a line chart and bar chart for exploring website traffic data.